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A Comparative Study of Vietnamese Embedding Models for LLM-Based Automated Question Answering on DUT Regulations

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***Abstract*—Question Answering (QA) is a specialized subfield within Information Retrieval (IR) that focuses on providing precise answers to user queries. Recent advances in Large Language Models (LLMs) have greatly improved the speed and accuracy of question-answering (QA) systems, leading to significantly enhanced capabilities and performance. This study introduces a multi-PDF querying system designed to support automated question answering for Vietnamese students, leveraging the capabilities of LLMs. To enhance the retrieval accuracy from natural language queries, we integrate four Vietnamese-language embedding models. These models are evaluated using a dataset comprising questions related to the academic regulations of the University of Da Nang – University of Science and Technology (DUT), reflecting real-life scenarios students commonly encounter. Experimental results reveal that the top-performing Vietnamese embedding model achieves a Mean Average Precision (mAP) of 0.840 and an F1-score of 0.959, demonstrating its high effectiveness in information retrieval. This study makes significant contributions to the integration of Vietnamese embedding models into automated question-answering systems, aiming to effectively support students in the university environment.**

Keywords—vietnamese embedding models, retrieval augmented generation, large language models, questing answering, informative retrieval

I. Introduction

Question-answering (QA) systems [1,2] represent a specialized area within Natural Language Processing (NLP) that focuses on generating accurate and contextually relevant answers to user questions, rather than simply retrieving documents. Recent research has increasingly focused on employing Large Language Models (LLMs) to enhance QA capabilities [3]. However, the deployment of such models in the Vietnamese language context remains limited due to linguistic and resource constraints.

Recent research has focused on building Vietnamese QA systems [4,5,6,7,8]. ViGPTQA [4], a practical question-answering system for Vietnamese built on large language models (LLMs), addressing the limited availability of resources for low-resource languages, was introduced. Minh-Nam Tran et al. tackled the shortage of Vietnamese abstractive QA datasets in the medical domain by introducing ViMedAQA [5], a dedicated dataset encompassing four key areas: body parts, diseases, drugs, and medicine. ViMedAQA is designed to evaluate the performance of LMMs in Vietnamese medical QA tasks. Experimental results highlight the models' capabilities in reasoning, memorization, and recognizing essential medical information within this domain.

In addition, extensive research efforts have been devoted to the development of QA systems leveraging Vietnamese legal documents [6,7]. In [6], the authors developed of a Vietnamese QA system specifically tailored to the legal domain, particularly the 2005 Law on Enterprises. The authors leverage existing Vietnamese NLP tools such as VietTokenizer and jvnTagger, as well as Lucene for information retrieval, and adapt cross-lingual algorithms like KEA for keyphrase extraction. Despite limited resources, the work aims to contribute to the foundation of Vietnamese QA systems by integrating both local and cross-language technologies. Thi-Hai-Yen Vuong et al. In [7], the authors introduced a Vietnamese legal QA system that utilizes article-level retrieval to tackle the complexity of legal texts and the shortage of marked data in low-resource languages. To address these challenges, the authors introduce a novel weak labeling method to enhance data quality and improve language model performance. The experimental results demonstrate that the proposed method effectively improves QA accuracy in Vietnamese legal contexts. Ba Thiem Nguyen et al. have proposed three main contributions: enhanced Vietnamese data processing techniques, an improved Reciprocal Rank Fusion method with normalized ordering, and a novel re-ranking strategy called Active Retrieval to refine information fed to large language models [8]. By integrating these innovations, the study presents a more accurate and stable QA system, effectively improving answer reliability in Vietnamese QA tasks.

Moreover, Trang Nguyen Thi Mai and Maxim Shcherbakov have investigated the use of BERT-based models for building a Vietnamese QA system using natural language processing (NLP) techniques [9]. The authors evaluated both multilingual BERT variants and the Vietnamese-specific PhoBERT model, finding that the monolingual PhoBERT model delivers superior performance. However, they also suggest that the multilingual BERT fine-tuned on XQuAD serves as an effective alternative when using multilingual models for Vietnamese QA tasks.

- Development of a QA system based on the Retrieval-Augmented Generation (RAG) framework, enabling precise and context-aware answers from Vietnamese-language content.

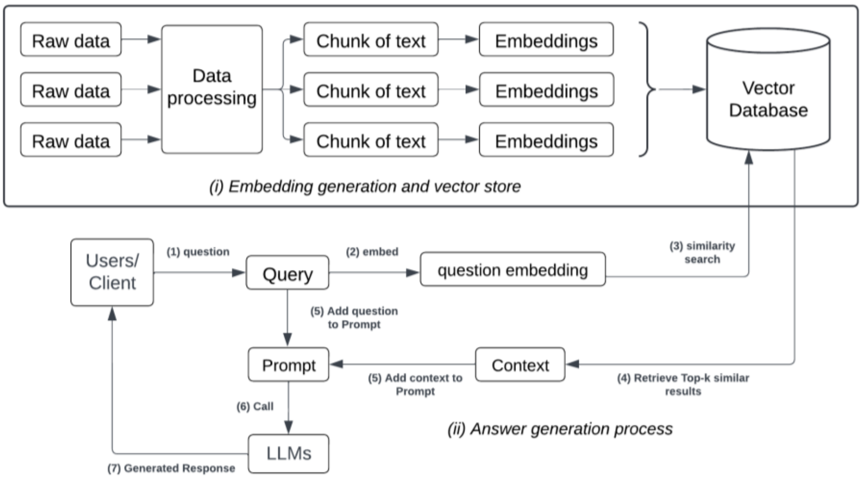

Fig. 1: The two parts of the QA system, (i) Embedding generation and vector store, (ii) Answer generation process

The application of QA systems in the context of a Vietnamese university environment brings unique challenges and opportunities. Vietnamese language processing requires specialized semantic embedding models due to the distinctive linguistic features of Vietnamese. Current QA systems mainly focus on languages with richer resources, such as English, leaving a gap in providing effective QA solutions for Vietnamese. The QA system for Vietnamese students consists of three core components: question query, information retrieval, and answer generation.

This study proposes an automated question-answering system that leverages Vietnamese embedding models to improve the accuracy and contextual relevance of answers for Vietnamese students. The system is applied in the context of processing Vietnamese-language content, specifically the policies and regulations of the University of Science and Technology – The University of Danang, to automatically respond to common student inquiries.

To evaluate system effectiveness, four open-source Vietnamese embedding models are integrated and compared through three key stages: question querying, information retrieval, and answer generation. Experimental results help identify the most optimal embedding model for QA systems within the context of higher education in Vietnam.

To the best of our knowledge, this is the first study to explore this specific area, with no existing dataset currently available for the problem. Therefore, this research contributes not only to the development of practical learning support tools for Vietnamese students but also paves the way for new research directions for Vietnamese Natural Language Processing.

The contributions of this research are threefold:

- Construction of a Vietnamese QA dataset for question-answering tasks, incorporating both GPT-generated and human-generated, students' questions that reflect common scenarios in the university environment.
- Evaluation and selection of optimal Vietnamese embedding models, identifying the most effective model for enhancing information retrieval and answer generation within the QA system.

The rest of the paper is organized as follows: Section II describes the proposed methodology; Section III reports and analyzes the experimental results; and Section IV concludes with a summary of the findings and suggestions for future research directions.

II. METHODOLOGY

A. *Question Answering System*

The QA system is built based on a retrieval-augmented generation technique combined with large language models. In our work, the QA system is designed to deliver accurate and comprehensive responses to student's inquiries based on the regulations of DUT, supporting them during their time at the university. This QA system consists of two main parts, as shown in Fig. 1: (i) Embedding generation and vector store, and (ii) Answer generation process.

In the first part, the documents are analyzed and divided into small texts, referred to as "Chunk of Text". Each chunk of text is then converted into vectors through pre-trained Vietnamese embedding models [10,11,12,13]. The detailed steps of this part are described as follows:

- **Data processing:** The raw data is preprocessed to remove noise.
- ***Text segmentation***: Text data are divided into a chunk of text.
- ***Embeddings extraction***: Each of the text segments is converted into an embedding vector through the embedding model.
- ***Embeddings store***: All embedding vectors are stored within the vector database, enabling efficient similarity search and retrieval during the question-answering process.

The second stage of this QA system is Handles query processing and answer generation. Specifically, the detailed steps of this part include:

- ***Question***: User query input.
- ***Embedding query questions***: The query is converted into a vector embedding (question embedding). This vector is extracted using the same embedding models.
- ***Semantic similarity search***: The output's result is a list of related text segments sorted in descending order of semantic similarity.
- ***Retrieve Top-k results***: The system selects the top k text segments that are most semantically similar to the input query.
- ***Construct prompt***: The query and the context are combined to form a prompt.
- ***Call LLM***: The prompt is fed into the LLM model.
- ***Generate response***: The LLM model generates an answer by leveraging both input query and the contextual information extracted from the retrieved text segments.

TABLE I. THE DETAILS OF EMBEDDING MODELS EVALUATED IN THIS WORK

Model Name	Source	#Params	Memory usage	Embedding Dimensions	Max Token
kvisbert [10]	keepitreal	135.00M	0.53 GB	768	258
mvibien [12]	maiduchuy32	135.00M	0.53 GB	768	258
stpmumv2 [13]	sentence-transformers	117.65M	0.46 GB	384	512
imule5b [14]	intfloat	278.04M	1.09 GB	768	514

B. Models

This work aims to evaluate well-known embedding models for Vietnamese languages [10,11,12,13,14]. These models are applied to the development of automated QA systems optimized for Vietnamese language processing. Each model is carefully assessed for its effectiveness in handling various Vietnamese language tasks, with a particular focus on sentence similarity. In this study, we employ three Vietnamese-adapted extensions of the Sentence-BERT (SBERT) model [11]: kvisbert [10], mvibien (a fine-tuned version of vietnamese-bi-encoder [12]), and stpmumv2 [13]. These models are specifically optimized for Vietnamese sentence similarity and information retrieval tasks, enabling efficient handling of complex queries with minimal memory and computational overhead. Among them, mvibien is a 135M-parameter embedding model fine-tuned for legal chatbot applications, making it particularly effective for question-answering systems in the legal domain by enhancing both accuracy and relevance. Stpmumv2 [13], on the other hand, is designed for multilingual tasks and includes Vietnamese support. Its small-size architecture allows for efficient deployment in resource-constrained settings while maintaining strong performance in sentence similarity tasks. In addition, we assess imule5b [14], a multilingual model supporting over 90 languages, including Vietnamese. It shows competitive performance in sentence similarity and embedding generation tasks. However, its performance may be reduced for low-resource languages.

Table I presents a comparison of these four different Vietnamese embedding models, highlighting their key specifications, including sources of model, number of parameters, memory usage, embedding dimensions, and max token. The *kvisbert* and *mvibien* models share identical configurations. Meanwhile, *stpmumv2* is designed to be more lightweight, making it suitable for resource-constrained environments. In addition, *imule5b* is the most complex and resource-intensive model.

C. Evaluation Metrics

In this study, we utilize standard evaluation metrics widely adopted in question-answering systems, including Mean Cosine Similarity, Precision, Recall, F1-score, Mean Average Precision (mAP), and Mean Reciprocal Rank (MRR).

Mean Cosine Similarity (mCS) is the similarity measure between the query vector and the corresponding retrieved vector, as illustrated in Equation 1.

$$mCS = \frac{1}{N}\sum_{i=0}^{N}\left(1 - \frac{Q_i \cdot D_i}{\|Q_i\|\|D_i\|}\right) \quad (1)$$

where Q_i is the query vector, and D_i is the document vector for Q_i. The score ranges from 0 to 1.

Precision quantifies the ratio of correctly retrieved answers to the total number of answers returned by the embedding model, reflecting its exactness. Recall, on the other hand, measures the proportion of correctly retrieved answers in relation to the total number of relevant answers, indicating the model's completeness. The F1-score, calculated as the harmonic mean of Precision and Recall, provides a balanced and comprehensive evaluation of the model's overall performance. These evaluation metrics are formally defined in Equation 2, 3, and 4.

$$Precision = \frac{number\ of\ correct\ answers}{Number\ of\ question\ answered} \quad (2)$$

$$Recall = \frac{number\ of\ correct\ answers}{number\ of\ questions\ to\ be\ answered} \quad (3)$$

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

Mean Average Precision (mAP) is a metric used to evaluate the precision of ranked retrieval results across multiple queries. It is computed as the average of the individual Average Precision (AP) scores for each query, as defined in Equation 5.

$$mAP = \frac{1}{N}\sum_{i=0}^{N} AP_i \quad (5)$$

where AP_i is the average precision for query i, computed as the mean of the precision values at each relevant result's rank, and N is the number of queries.

Mean Reciprocal Rank (mRR) [15] is a widely used metric for evaluating the quality of ranked results in information retrieval systems. It is calculated as the average of the reciprocal ranks of the first correctly retrieved relevant text segment for each query question, providing insight into how highly relevant results are ranked. This metric is formally defined in Equation 6.

$$MRR = \frac{1}{N}\sum_{i=0}^{N}\frac{1}{r_i} \quad (6)$$

where r_i is the rank of the first relevant text segment for i^{th} query, and N is the number of queries.

D. Dataset

The raw textual data was collected from the official website of the University of Da Nang – University of Science and Technology (DUT). Documents containing the most recent and officially published university regulations were collected. To ensure data quality, the documents were preprocessed by removing irrelevant content, redundant formatting, and textual noise. Finally, all processed data is combined into a single PDF file for use in downstream tasks.

To create a robust QA dataset, we employed GPT-4 [16] to generate question–answer pairs from the curated document corpus. Specifically, 15 question–answer pairs were generated using GPT-4 based on selected paragraphs, ensuring coverage of various regulatory themes such as graduation requirements, capstone projects, foreign language proficiency standards, and academic training programs. Additionally, 35 context-based questions were manually designed by reviewing the documents and formulating realistic scenarios that DUT students may encounter in academic and administrative contexts. This hybrid approach ensures both linguistic diversity and contextual relevance in the QA dataset.

E. Integrating Gemini LLM for contextual Answering

The LLM integrated into our QA system is Gemini-1.5-flash-lastest, a state-of-the-art LLM [17] designed to optimize query processing based on contextual information. In our system, individual text segments are first combined into a coherent paragraph. Gemini-1.5-Flash then leverages this structured context, along with the user query, to procedure accurate and contextually appropriate responses, thereby ensuring high-quality and coherent answers.

III. EXPERIMENTAL RESULTS

This section provides a comparative analysis of four embedding models, assessed using key evaluation metrics: mCS, mAP, and mRR. The results are summarized in Table II. Additionally, Precision, Recall, and F1-score are employed to further evaluate model performance, as shown in Table III. In both tables, the highest scores are highlighted in bold, while the lowest scores are underlined for ease of interpretation.

TABLE II. COMPARATIVE PERFORMANCE OF THE FOUR EMBEDDING MODELS

Models	mCS	mAP	mRR
kvisbert [10]	0.530	0.625	0.636
mvibien [12]	0.527	**0.840**	**0.841**
stpmumv2 [13]	**0.532**	0.274	0.274
imule5b [14]	0.163	0.657	0.665

Table II shows that the *mvibien* model achieves the highest performance in both the mAP and mRR, with scores of 0.840 and 0.841, respectively. These results indicate its superior ability to accurately retrieve and rank relevant information. In contrast, the *stpmumv2* model gives the lowest scores for both mAP (0.274) and mRR (0.274), indicating significant limitations in both retrieving relavant information and ranking capability. With respect to Mean Cosine Similarity (mCS), the *stpmumv2* model achieves the highest score of 0.532, demonstrating strong capability in retrieving semantically similar results in vector space. Conversely, *imule5b* performs the poorest in this metric, with an mCS score of 0.163, indicating its relative ineffectiveness in capturing similarity between query and candidate embeddings. However, it shows relatively strong performance in mAP (0.657) and mRR (0.665), suggesting that it can retrieve relevant answers, even if the vector similarity isn't as strong. Although *stpmumv2* achieves relatively average in mCS (0.532), it lags in retrieval accuracy (mAP: 0.274, mRR: 0.274), suggesting its embeddings may not be sufficiently optimized for ranking relevance.

Overall, *mvibien* stands out for its exceptional retrieval precision and ranking performance. It achieved the highest results in both the mAP and mRR scores, making it the most reliable in terms of ranking and retrieving relevant information. In contrast, *stpmumv2* is the weakest model with the lowest score on the board. Although *stpmumv2* achieves the highest score in mCS, it is overshadowed by *mvibien* in the other two criteria. Although *imule5b* achieves a low score in mCS, it shows high potential in the rankings in mAP and mRR scores.

TABLE III. THE COMPARATIVE RESULTS OF VIETNAMESE EMBEDDING MODELS IN RAG WITH GEMINI-1.5-FLASH-LATEST

Models	Precision	Recall	F1-score
kvisbert [10]	0.871	0.780	0.823
mvibien [12]	**0.958**	**0.960**	**0.959**
stpmumv2 [13]	0.950	0.400	0.563
imule5b [14]	0.950	0.800	0.869,

In Table III, *mvibien* achieves the best performance with almost perfect scores in all three metrics: Precision at 0.958, Recall at 0.960, and F1-score at 0.959. On the contrary, *stpmumv2* is the weakest model with significantly low scores, especially in Recall (0.400) and F1-score (0.563). *Mvibien, imule5b*, and *stpmumv2* lead in Precision with scores of 0.958, 0.950 and 0.950. This shows that the above models achieve accurate results with the most relevant results out of the total results that the model gives. In terms of Recall, the *mvibien* model demonstrates superior performance, achieving the highest recall score of 0.960. This indicates that it is highly effective in retrieving relevant answers relative to the total number of questions, reflecting near-complete coverage. The *imule5b* and *kvisbert* models follow, with recall scores of 0.800 and 0.780, respectively. In contrast, *stpmumv2* records a substantially lower recall of 0.400, suggesting a limited capacity to retrieve the full range of relevant answers, despite exhibiting high precision in its outputs. With respect to the F1-score, the *mvibien* model achieves the highest performance, attaining a score of 0.959. The *imule5b* model ranks second with an F1-score of 0.869,

TABLE IV. THE RESPONSES GENERATED BY GEMINI-1.5-FLASH-LATEST USING THE KVISBERT AND MVIBIEN MODEL

Question	Document	Kvisbert Response	AP - RR	Mvibien Response	AP - RR
cách tính điểm trung bình	*Cách tính điểm trung bình chung. Điểm trung bình chung học kỳ và điểm trung bình chung tích luỹ được tính theo công thức sau và được làm tròn đến 2 chữ số thập phân:* $A = \frac{\sum_{i=1}^{n}(a_i.n_i)}{\sum_{i=1}^{n} n_i}$ *Trong đó:* *A là điểm trung bình chung học kỳ hoặc điểm trung bình chung tích lũy.* a_i *là điểm của học phần thứ i* n_i *là số tín chỉ của học phần thứ i* *n là tổng số học phần.* *Điểm trung bình chung học kỳ để xét học bổng, khen thưởng sau mỗi học kỳ chỉ tính theo điểm của học phần ở lần học thứ nhất.* *Điểm trung bình chung học kỳ và điểm trung bình chung tích lũy để xét thôi học, xếp hạng học lực sinh viên và xếp hạng tốt nghiệp được tính theo điểm cao nhất trong các lần học.*	*Bài viết này cung cấp thông tin về cách tính điểm trung bình chung học kỳ và điểm trung bình chung tích lũy. Công thức tính điểm trung bình chung được cho là:* *Σ(ai * ni) / Σni* *Trong đó:* ** A là điểm trung bình chung học kỳ hoặc điểm trung bình chung tích lũy* ** ai là điểm của học phần thứ i* ** ni là số tín chỉ của học phần thứ i* ** n là tổng số học phần* *Tuy nhiên, bài viết không cung cấp thông tin chi tiết về cách tính điểm trung bình cụ thể cho mỗi học phần.*	*1 - 1*	*Bài viết cung cấp thông tin về cách tính điểm trung bình chung học kỳ và điểm trung bình chung tích lũy. Công thức tính là:* *Σ(ai * ni) / Σni* *Trong đó:* ** A là điểm trung bình chung học kỳ hoặc điểm trung bình chung tích lũy* ** ai là điểm của học phần thứ i* ** ni là số tín chỉ của học phần thứ i* ** n là tổng số học phần.* *Bài viết cũng có giải thích cách tính điểm trung bình chung học kỳ để xét học bổng, khen thưởng, thôi học, xếp hạng học lực sinh viên và xếp hạng tốt nghiệp.*	*1 - 1*
ĐATN cần được thực hiện trong bao lâu?	*Thời gian làm ĐATN mỗi đợt là 15 tuần*	*Theo văn bản bạn cung cấp, mỗi đợt thực hiện ĐATN là **15 tuần**.*	*0.16 – 0.16*	*Theo văn bản bạn cung cấp, mỗi đợt thực hiện ĐATN là **15 tuần**.*	*1 – 1*
Sinh viên sẽ bị cảnh báo kết quả học tập khi nào	*Việc cảnh báo kết quả học tập của sinh viên dựa trên các điều kiện sau:* *a) Có điểm trung bình chung tích lũy đạt dưới 1,20 đối với sinh viên năm thứ nhất (theo xếp hạng năm đào tạo); dưới 1,40 đối với sinh viên năm thứ hai; dưới 1,60 đối với sinh viên năm thứ ba hoặc dưới 1,80 đối với sinh viên các năm tiếp theo và cuối khoá;* *b) Có điểm trung bình chung học kỳ đạt dưới 0,80 đối với học kỳ đầu của khóa học; đạt dưới 1,00 đối với các học kỳ tiếp theo;*	*Sinh viên sẽ bị cảnh báo kết quả học tập sau mỗi học kỳ dựa trên điểm trung bình chung tích lũy và điểm trung bình chung học kỳ.*	*1 - 1*	*Sinh viên sẽ bị cảnh báo kết quả học tập khi có điểm trung bình chung tích lũy đạt dưới 1,20 đối với sinh viên năm thứ nhất, dưới 1,40 đối với sinh viên năm thứ hai, dưới 1,60 đối với sinh viên năm thứ ba hoặc dưới 1,80 đối với sinh viên các năm tiếp theo và cuối khóa; hoặc có điểm trung bình chung học kỳ đạt dưới 0,80 đối với học kỳ đầu của khóa học; đạt dưới 1,00 đối với các học kỳ tiếp theo.*	*1 - 1*
sinh viên được làm đồ án tốt nghiệm cần đủ những điều kiện gì	*Sinh viên được làm đồ án tốt nghiệp khi có đủ các điều kiện sau đây:* *a) Đã học tất cả các học phần (trừ Đồ án tốt nghiệp) của chương trình đào tạo, tích lũy đủ các học phần tiên quyết của đồ án tốt nghiệp và khối lượng kiến thức còn chưa tích lũy không vượt quá 4 tín chỉ.* *b) Điểm trung bình chung tích luỹ tính từ đầu khóa học không dưới 1,90.*	*Văn bản không đề cập đến điều kiện để làm đồ án tốt nghiệp.*	*0 – 0*	*Theo văn bản, sinh viên được làm đồ án tốt nghiệp khi có đủ các điều kiện sau:* ** Đã học tất cả các học phần (trừ Đồ án tốt nghiệp) của chương trình đào tạo, tích lũy đủ các học phần tiên quyết của đồ án tốt nghiệp và khối lượng kiến thức còn chưa tích lũy không vượt quá 4 tín chỉ.* ** Điểm trung bình chung tích lũy tính từ đầu khóa học không dưới 1,90.* *Văn bản không đề cập thêm điều kiện nào khác để được làm đồ án tốt nghiệp.*	*1 – 1*

followed by *kvisbert* at 0.823, both reflecting relatively well-balanced retrieval capabilities. In contrast, *stpmumv2* records the lowest F1-score at 0.563, highlighting its limited overall effectiveness despite its high precision.

Table IV shows the responses generated by Gemini-1.5-flash-latest for two models, *kvisbert* and *mvibien*. In the first question, both models provide correct answers, indicating their accuracy in retrieval. Moving on to the second question, although both models produce correct answers, *mvibien* achieved significantly higher AP and RR scores compared to *kvisbert*. For the third question, despite having the same AP and RR scores, *mvibien* gives a more accurate and complete answer, whereas *kvisbert*'s answer is general and lacks detail. Finally, in the last question, while *mvibien* extracts and answers the question correctly, *kvisbert* can not extract the answer. This table demonstrates *mvibien*'s superiority in extracting information from the given document.

IV. CONCLUSION

This work focuses on identifying the most effective Vietnamese embedding model for integration into an automated QA system specifically designed for students at the University of Da Nang – University of Science and Technology. By simulating real-life scenarios that DUT students may encounter, we construct a specialized QA dataset and develop a system based on the RAG architecture. Four Vietnamese-language embedding models are selected, integrated, and systematically evaluated using standard performance metrics. Among the models assessed, mvibien demonstrates superior performance across multiple evaluation criteria. As a result, it stands out as the most effective embedding model for enhancing the accuracy and overall performance of the proposed QA system on the dataset related to university regulations.

Future work will focus on refining the RAG architecture by incorporating metadata into the retrieval process. This extension is expected to improve retrieval precision and contextual relevance, ultimately leading to more accurate and informative responses for users. Moreover, the current hybrid QA dataset comprises 50 questions, integrating automatically generated items with manually crafted, context-specific questions adapted to the academic and administrative needs of DUT students. While this initial dataset provides a valuable foundation, its limited size may restrict the statistical robustness and generalizability of Vietnamese embedding models. To enhance model performance and ensure broader applicability, future work will focus on significantly expanding the dataset to include a wider array of regulated academic topics and diverse linguistic styles. This expansion will facilitate more precise testing and fine-tuning of the QA system, improving its accuracy and effectiveness as a practical tool for student support.

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